Area and volume — Part 1

Geometry and Trigonometry · Math · 55 questions

1.1 c53570c0-9d6b-49d6-8b3f-c45a4ef532db · Easy

The perimeter of triangle *ABC* is 17 inches, the length of side *AB* is 4 inches, and the length of side *AC* is 7 inches. What is the length, in inches, of side *BC*?

A. 4

B. 6

C. 7

D. 11

1.2 7fb7e2c6-237c-42b9-a178-4810e8c6cd93 · Easy

A rectangle has a length of 3 units and a width of 39 units. Which expression gives the area, in square units, of this rectangle?

A. $23 + 39$

B. $23 \cdot 39$

C. $3 + 39$

D. $3 \cdot 39$

1.3 5d25f220-0193-4939-a687-673246d3076b · Easy

Rectangle *ABCD* has an area of 280 square centimeters. The length of the rectangle is 10 centimeters. What is the width, in centimeters, of the rectangle?

A. 10

B. 28

C. 76

D. 140

1.4 cdf507c8-4e6e-437e-bae9-d9f6a9e92f13 · Easy

A right rectangular prism has a length of 11 meters, a width of 8 meters, and a height of 10 meters. What is the volume, in cubic meters, of the prism?

Student-produced response: _____

1.5 cde76af2-5837-4a16-aae2-109dd079e384 · Easy

What is the perimeter, in inches, of a rectangle with a length of 4 inches and a width of 9 inches?

A. 13

B. 17

C. 22

D. 26

1.6 a3a441b7-c6ac-48e3-9e5c-759686bbc5ee · Easy

The length of one side of an equilateral triangle is 28 inches. What is the perimeter, in inches, of this triangle?

A. 28

B. 31

C. 84

D. 112

1.7 2bde9f21-d49c-42af-b7ec-38c87b242d02 · Easy

The area of a rectangle is 57 square inches. The length of the longest side of the rectangle is 19 inches. What is the length, in inches, of the shortest side of this rectangle?

Student-produced response: _____

1.8 5a173e5b-ac90-4f05-806e-fe38d80fcfd9 · Easy

A rectangle has a length of 56 inches and a width of 28 inches. What is the area, in square inches, of the rectangle?

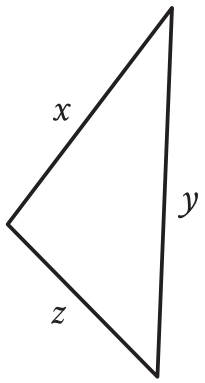
A. 28

B. 84

C. 168

D. $1{,}568$

1.9 2fcb111d-48d0-4ee3-925e-0ffeec3615a4 · Easy



Note: Figure not drawn to scale.

- The lengths of the sides are x, y, and z.
- A note indicates the figure is not drawn to scale.

The triangle shown has a perimeter of 22 units. If $x = 9$ units and $y = 7$ units, what is the value of z, in units?

A. 6

B. 7

C. 9

D. 16

1.10 9a1d9f4a-bbb7-4e9d-bb0c-c5445263d935 · Easy

The length of the base of a triangle is 8 inches, and the height of the triangle is 12 inches. What is the area, in square inches, of the triangle?

A. 20

B. 48

C. 60

D. 96

1.11 4a39e52e-02ea-49d7-a695-87a667bdb785 · Easy

The area of a square is 64 square inches. What is the side length, in inches, of this square?

A. 8

B. 16

C. 64

D. 128

1.12 05f34b6c-f496-4574-bc7b-cffaaa469975 · Easy

What is the area, in square inches, of a rectangle with a length of 7 inches and a width of 6 inches?

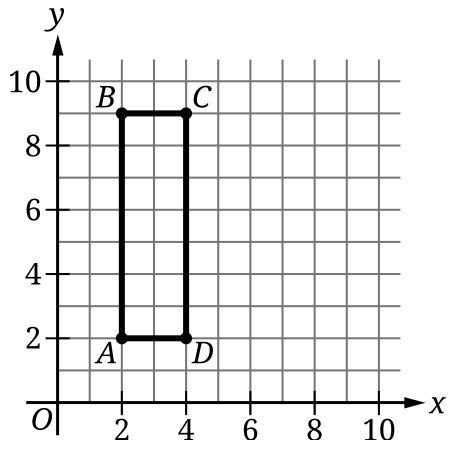
A. 13

B. 20

C. 42

D. 84

1.13 21882da9-ffd9-4340-8877-b77ba9972851 · Easy



- Rectangle upper A upper B upper C upper D has vertices at the following points:
  - upper A: (2 comma 2)
  - upper B: (2 comma 9)
  - upper C: (4 comma 9)
  - upper D: (4 comma 2)

Rectangle *ABCD* shown has a length of 7 units and a width of 2 units. What is the area, in square units, of rectangle *ABCD*?

A. 5

B. 9

C. 14

D. 28

1.14 7ce3a2e8-bebb-4040-84e7-3f30170327e3 · Easy

A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?

Student-produced response: _____

1.15 c2922131-fa53-47c8-b19a-4f809b49ffe3 · Easy

What is the area, in square centimeters, of a rectangle with a length of 34 centimeters cm and a width of 29 cm?

Student-produced response: _____

1.16 77d41ee0-634a-4578-a6b0-6ec8687d2c0a · Easy

Rectangle P has an area of 72 square inches. If a rectangle with an area of 20 square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

A. 92

B. 84

C. 80

D. 52

1.17 025774-DC · Easy

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $DE = 2AB$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

A. 10

B. 40

C. 80

D. 120

1.18 f5e0aefd-1908-42ec-9072-52390bef3063 · Easy

What is the area, in square centimeters, of a rectangle with a length of 36 centimeters and a width of 34 centimeters?

A. 70

B. 140

C. $1{,}156$

D. $1{,}224$

1.19 76f07394-0a95-4e65-86b8-e42618d65c92 · Easy

What is the area of a rectangle with a length of 4 centimeters cm and a width of 2 cm?

A. 6 cm^2

B. 8 cm^2

C. 12 cm^2

D. 36 cm^2

1.20 21b61a47-8395-41b7-9031-ab756ca63c22 · Easy

The side length of a square is 55 centimeters cm. What is the area, in cm^2, of the square?

A. 110

B. 220

C. $3{,}025$

D. $12{,}100$

1.21 1e523d74-12a9-4f3a-8279-13ec77276c13 · Easy

Each side of a square has a length of 45. What is the perimeter of this square?

Student-produced response: _____

1.22 8ec98031-aeed-4459-a154-8c48f0ec122a · Easy

The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

A. 145

B. 113

C. 82

D. 81

1.23 786b1f64-6764-44c0-a7d6-75880720fc77 · Easy

Each base of a right rectangular prism has a length of 19 inches and a width of 8 inches. The prism has a volume of $2{,}736$ cubic inches. What is the height, in inches, of the prism?

A. 18

B. 27

C. 144

D. 152

1.24 ca7c086a-76e8-4959-b23f-94d448a1c1c3 · Easy

The width of a rectangle is 7 centimeters. The length of the rectangle is 40 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

A. 7

B. 14

C. 54

D. 329

1.25 f06ed67a-28a2-4bfe-9834-47786c3571e8 · Easy

The lengths of two sides of a triangle are 4 centimeters and 6 centimeters. If the perimeter of the triangle is 18 centimeters, what is the length, in centimeters, of the third side of this triangle?

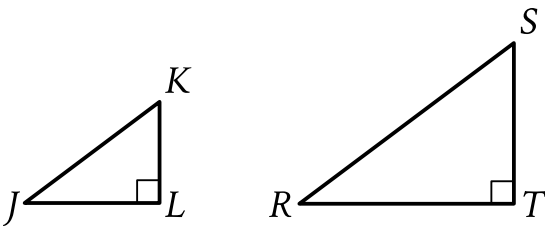
A. 2

B. 8

C. 10

D. 24

1.26 b2842dd1-e205-4ff0-a977-6379b5655857 · Easy



Note: Figure not drawn to scale.

- Right triangle upper J upper K upper L is labeled as follows:
  - Angle upper L is a right angle.
- Right triangle upper R upper S upper T is labeled as follows:
  - Angle upper T is a right angle.
- A note indicates the figures are not drawn to scale.

In the figure shown, triangle *JKL* is similar to triangle *RST*, where *J* corresponds to *R* and *K* corresponds to *S*. The length of $\bar{JK}$ is 15, and the perimeter of triangle *JKL* is 36. The length of $\bar{RS}$ is 135. What is the perimeter of triangle *RST*?

A. 324

B. 540

C. $2{,}916$

D. $8{,}100$

1.27 09ef21b2-86ef-4977-8eef-37b47c81624e · Easy

Triangle R has an area of 80 square centimeters cm^2. Square S has side lengths of 4 cm. What is the total area of triangle R and square S, in cm^2?

A. 42

B. 44

C. 84

D. 96

1.28 8cb6a96f-5086-4d10-8015-3085accf0ee2 · Easy

Which expression represents the area, in square inches, of a square with a side length of 35 inches?

A. $\sqrt{35}$

B. $35 + 35$

C. $(4)(35)$

D. 35^2

1.29 60500dd8-893a-4fea-903d-de896120774c · Easy

A rectangle has a length of 13 and a width of 6. What is the perimeter of the rectangle?

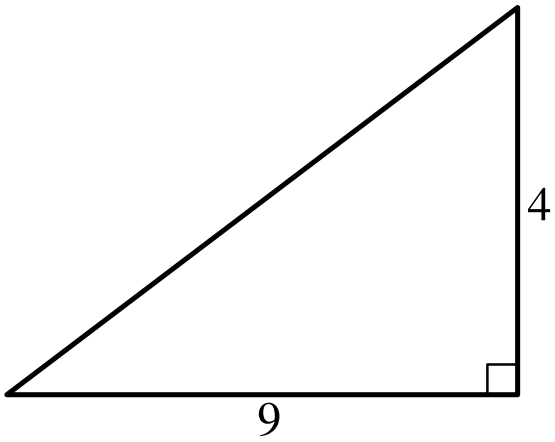
A. 12

B. 26

C. 38

D. 52

1.30 cab19268-2903-44cd-aa02-6f253b7a4efc · Easy



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the sides adjacent to the right angle are as follows:
  - 4
  - 9
- A note indicates the figure is not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

A. 18

B. 13

C. 9

D. 4

1.31 0c4e91ab-c388-4002-8b80-16bf6f098d01 · Easy

A rectangle has an area of 63 square meters and a length of 9 meters. What is the width, in meters, of the rectangle?

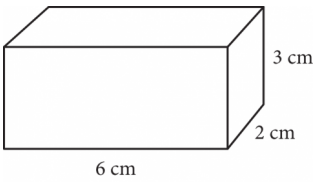
A. 7

B. 54

C. 81

D. 567

1.32 027084-DC · Easy



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh, where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

A. 36

B. 24

C. 12

D. 11

1.33 a5cc3d06-5c10-40d6-bfec-b01230ec82b8 · Easy

What is the area of a rectangle with a length of 17 centimeters cm and a width of 7 cm?

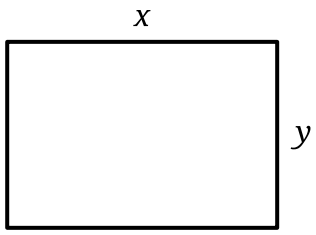
A. 24 cm^2

B. 48 cm^2

C. 119 cm^2

D. 576 cm^2

1.34 f9aa6eb4-a501-4825-9c17-1efda9ecd0cf · Easy



Note: Figure not drawn to scale.

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

A. 7

B. 12

C. 35

D. 40

1.35 7b4cffc9-9565-413b-810d-67d7f92aa56e · Easy

A triangle has a base length of 40 centimeters and a height of 90 centimeters. What is the area, in square centimeters, of the triangle?

Student-produced response: _____

1.36 2cc3f845-5222-4c22-8764-9d19c7b188fb · Easy

The area of a rectangle is 630 square inches. The length of the rectangle is 70 inches. What is the width, in inches, of this rectangle?

A. 9

B. 70

C. 315

D. 560

1.37 13e26f81-104e-4cc5-b4b5-52d4234d4b53 · Medium

The area of a triangle is 36 square units. If the height of this triangle is 2 units, what is the length, in units, of the base of this triangle?

Student-produced response: _____

1.38 bf7a641a-e835-4a95-a957-157c3c91e4ca · Medium

Square X has a side length of 12 centimeters. The perimeter of square Y is 2 times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

A. 6

B. 10

C. 14

D. 24

1.39 9f8442fc-4e43-48a4-9eda-fd446fd06043 · Medium

A right circular cylinder has a volume of 377 cubic centimeters. The area of the base of the cylinder is 13 square centimeters. What is the height, in centimeters, of the cylinder?

Student-produced response: _____

1.40 2576e4c1-c4ee-4f32-8350-159f08967b70 · Medium

Square A has side lengths that are 246 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k?

A. $60{,}516$

B. 492

C. 246

D. 123

1.41 f59b253c-f13a-4027-850d-9d1abc35bde5 · Medium

The length of each edge of a box is 29 inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

Student-produced response: _____

1.42 029396-DC · Medium

What is the length of one side of a square that has the same area as a circle with radius 2 ?

A. 2

B. $\sqrt{2\pi}$

C. $2\sqrt{\pi}$

D. 2π

1.43 7ffcadaa-3611-4602-92ad-c14085a27aef · Medium

A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b?

Student-produced response: _____

1.44 357b0938-dcf1-4984-934a-7d55d036717a · Medium

A triangular prism has a height of 8 centimeters cm and a volume of 216 cm^3. What is the area, in cm^2, of the base of the prism? (The volume of a triangular prism is equal to Bh, where B is the area of the base and h is the height of the prism.)

Student-produced response: _____

1.45 7c5d80a9-e593-4cf8-b36b-bd89ed6226c7 · Medium

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

A. 700

B. 350

C. 175

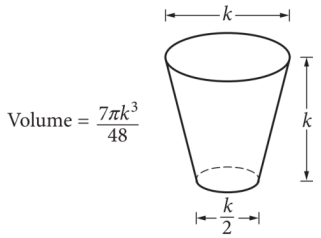
D. 80

1.46 86cc8eb8-5cbf-4188-9abd-1839bfec0526 · Medium

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 2 centimeters. The height of this pyramid is 7 centimeters. What is the volume, in cubic centimeters, of this pyramid?

Student-produced response: _____

1.47 070925-DC · Medium



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k, in centimeters?

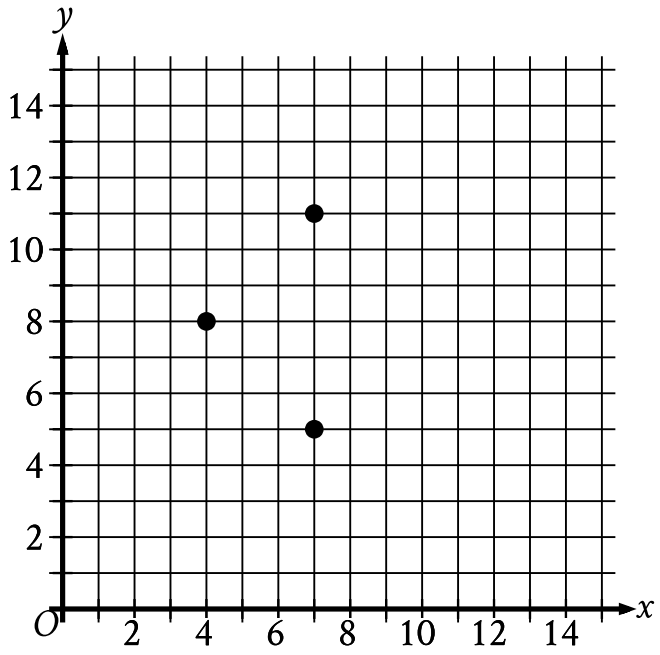
A. 2.52

B. 7.67

C. 7.79

D. 10.11

1.48 924fda1b-aa0e-4da6-bd05-7a55e96639dd · Medium



- From left to right, the 3 points have the following coordinates:
  - (4 comma 8)
  - (7 comma 11)
  - (7 comma 5)

The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k?

A. 3

B. 6

C. 7

D. 9

1.49 66a68c59-a01d-4c33-b54e-5a44e513c248 · Medium

Circle K has a radius of 4 millimeters mm. Circle L has an area of 100π mm^2. What is the total area, in mm^2, of circles K and L?

A. 14π

B. 28π

C. 56π

D. 116π

1.50 06614-DC · Medium

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

A. $16{,}000\,\pi$

B. $20{,}000\,\pi$

C. $25{,}000\,\pi$

D. $31{,}250\,\pi$

1.51 a6f2f98a-1452-4d23-aa35-d397a85c694f · Medium

The area of one of the bases of a right circular cylinder is 81 square centimeters, and the volume of the cylinder is $2{,}187$ cubic centimeters. What is the height, in centimeters, of the cylinder?

A. 27

B. $2{,}025$

C. $2{,}106$

D. $177{,}147$

1.52 22accec5-7fd4-4bbf-8af1-a6077fe56fc3 · Medium

Circle N has a radius of 7 millimeters mm. Circle M has an area of 64π mm^2. What is the total area, in mm^2, of circles N and M?

A. 113π

B. 92π

C. 78π

D. 15π

1.53 ea4fb9f5-47ef-4d74-89e2-5f8e44710904 · Medium

A triangle has a base length of 56 centimeters and a height of 112 centimeters. What is the area, in square centimeters, of the triangle?

A. 168

B. $1{,}568$

C. $3{,}136$

D. $6{,}272$

1.54 39d02567-fb2f-4a7c-a802-69186c6a894f · Medium

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

Student-produced response: _____

1.55 00113-DC · Medium

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of $75\ cm^2$, and the height of the can is 10 cm. If $110\ cm^3$ of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

A. $7.5\ cm^3$

B. $185\ cm^3$

C. $640\ cm^3$

D. $750\ cm^3$